POWER PULSE

Analog Meters Basics Important Formulae

Notes by Balaji Sir

1 Torques in an indicating instrument

1. $$T_d = T_c \ \ (\text{spring control: } T_c = K\theta)$$

steady deflection; T_d = deflecting torque; T_c = control torque

2. $$\omega_n = \sqrt{K/J}, \qquad \zeta = \frac{C}{2\sqrt{KJ}}$$

ζ < 1 under-damped; ζ = 1 critically damped; ζ > 1 over-damped; the aim is ζ ≈ 0.7

2 PMMC

3. $$T_d = B\,N\,A\,I, \qquad \theta = \frac{BNA}{K}\,I$$

deflection ∝ current: uniform scale; works on DC only

4. $$\text{With rectifier: form factor} = \frac{V_{rms}}{V_{avg}} = 1.11 \ \ (\text{sine}), \qquad \text{Reading} = 1.11 \times V_{avg}$$

the scale is calibrated in rms for a sine wave

3 Moving iron (MI)

5. $$T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}, \qquad \theta = \frac{I^2}{2K}\frac{dL}{d\theta}$$

square-law (non-uniform) scale; reads rms; works on AC and DC

4 Electrodynamometer

6. $$T_d = I_1 I_2 \frac{dM}{d\theta}\cos\phi, \qquad \theta \propto I^2 \ \ (\text{ammeter / voltmeter})$$

reads rms on AC and DC; as a wattmeter θ ∝ V I cos φ

5 Comparison

7. PMMC: DC, accurate MI: AC/DC, cheap Dynamometer: AC/DC, transfer instrument